CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 68 (2020) 101740

Contents lists available at ScienceDirect

Calphad

journal homepage: http://www.elsevier.com/locate/calphad

Thermodynamic assessment of the RE-B (RE = Ce, Dy, Lu) binary systems

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ARTICLE INFO

Keywords:
RE-B
Phase diagram
Thermodynamic
CALPHAD

ABSTRACT

Phase equilibria and thermodynamic properties of Nd-Fe-B-based alloys are fundamental to design novel Nd-Fe-B-based permanent magnets with outstanding magnetic properties and lower costs. In order to develop thermodynamic database of multi-component Nd-Fe-B-based alloys with the rare earth (RE) metals, in this work, the RE-B (RE = Ce, Dy, Lu) binary systems were assessed thermodynamically using the CALPHAD method based on the available phase diagram and thermodynamic data reported in the literature. In the thermodynamic modeling, the solution phases including liquid, bcc, fcc, hcp and rhombohedral, are described as the substitutional solution model and their excess Gibbs energies are described by Redlich-Kister polynomial. The binary intermetallic compounds, CeB_4, CeB_6, DyB_2, DyB_4, DyB_6, DyB_{12}, DyB_{66}, LuB_2, LuB_4, LuB_{12} and LuB_{66}, are treated as the stoichiometric compounds considering the experimental heat capacities and enthalpy increments of some intermetallic compounds. The calculated results are in good agreement with the reported experimental data.

1. Introduction

Nd-Fe-B sintered permanent magnets have been widely used in advanced wind generators, aerospace, transportation, communication and electric machinery industries because of their excellent magnetic properties [1–5]. The expensive heavy rare earth (RE) metals (e.g. Dy, Tb, Lu) are important alloying elements to enhance the coercivity and thermal stability of Nd-Fe-B permanent magnets. However, the great demand of Nd-Fe-B permanent magnets accelerates the consumption of heavy rare earth metals. Recently, partial substitution of cheap light rare earth metals (e.g. La, Ce, Y) for expensive heavy rare earth metals in Nd-Fe-B permanent magnets to reduce their costs was explored extensively and thus the microstructure and magnetic properties of Nd-Fe-B permanent magnets with light rare earth metals were investigated experimentally [6–10]. The effect of high-abundant rare earth metals on the microstructure and magnetic properties of Nd-Fe-B permanent magnets could be related to phase equilibria and phase formation of RE-Fe-B alloys. The knowledge of phase equilibria and thermodynamic information of multi-component RE-Fe-B alloys are indispensable to design alloy compositions and processes of Nd-Fe-B permanent magnets with excellent magnetic properties and low costs.

In order to develop thermodynamic database of the RE-Fe-B permanent magnetic alloys, the Fe-RE and RE-B binary systems as key binary systems are necessary. Many Fe-RE binary systems such as Fe–Pr, Fe–Nd, Fe–Dy, Fe–Tb, Fe–Gd, Fe–Sm, Fe–Ho, Fe–Tm, Fe–La, Fe–Ce and Fe–Y were assessed in our previous work [11–14] and in the literature [15–17], while only some RE-B binary systems (e.g. La–B, Ce–B, Pr–B and Nd–B) were optimized [18–22]. In particular, the Ce–B binary system was optimized by Wang et al. [21] and Premovic et al. [22] using the CALPHAD method. The calculated temperatures of invariant reactions in the Ce–B binary system by Wang et al. [21] and Premovic et al. [22] are in agreement with the experimental data [23,24]. However, the experimental heat capacities and enthalpy increments of two intermetallic compounds CeB_4 and CeB_6 were not used by Wang et al. [21], while the calculated results by Premovic et al. [22] indicates that the enthalpy of mixing of liquid phase in the Ce-rich part shows an abnormally positive and the inverted miscibility gap of liquid phase at high temperature (above 7000 K) is found. The Dy–B binary system was optimized by Ende et al. [25] using the modified quasi-chemical model to describe liquid phase. There is an obvious difference between the calculated temperature (2749.3 K) of the congruent reaction ($L \leftrightarrow DyB_4$) [25] and the estimated data (2773.0 K) assumed by Spear [23]. In addition, the Lu–B binary system was not assessed in the reported literature.

Therefore, thermodynamic assessment of the RE-B (RE = Ce, Dy, Lu) binary systems was performed in this work using the CALPHAD method

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https://doi.org/10.1016/j.calphad.2020.101740
Received 5 August 2019; Received in revised form 28 December 2019; Accepted 11 January 2020
Available online 21 January 2020
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Table 1
Crystallographic information of stable phases in the RE-B (RE = Ce, Dy, Lu) binary system.

Phase	Prototype	Space group	Pearson symbol	Reference
δ-Ce	Cu	$Im\bar{3}m$	cI2	[41]
γ-Ce	W	$Fm\bar{3}m$	cF4	[41]
CeB_4	ThB_4	P4/mbm	tP20	[41]
CeB_6	CaB_6	$Pm\bar{3}m$	cP7	[41]
β-Dy	W	$Im\bar{3}m$	cF4	[35]
α-Dy	Mg	$P6_3/mmc$	cI2	[35]
DyB_2	AlB_2	P6/mmm	hP3	[35]
DyB_4	ThB_4	P4/mbm	tP20	[36]
DyB_6	CaB_6	$Pm\bar{3}m$	cP7	[36]
DyB_{12}	UB_{12}	$Fm\bar{3}m$	cF52	[36]
DyB_{66}	YB_{66}	$Fm\bar{3}c$	cF1936	[36]
α-Lu	Mg	$P6_3/mmc$	cI2	[42]
LuB_2	AlB_2	P6/mmm	hP3	[42]
LuB_4	UB_4	P4/mbm	tP20	[42]
LuB_{12}	UB_{12}	$Fm\bar{3}m$	cF52	[42]
LuB_{66}	YB_{66}	$Fm\bar{3}c$	cF1936	[42]
β-B	β-B	$R\bar{3}m$	hR105	[35]

on the basis of the experimental information reported in the literature, which could provide thermodynamic parameters in the development of thermodynamic database of multi-component RE-Fe-B magnetic alloys.

2. Literature information

2.1. The Ce–B binary system

The phase diagram of the Ce–B binary system was determined experimentally by Spear [23] by means of differential thermal analysis (DTA), X-ray powder diffraction (XRD) and electron microscope. Two intermetallic compounds (CeB_4 and CeB_6) are stable in the Ce–B binary system. According to Spear [23], the Ce–B binary system has two eutectic reactions (L ↔ β-B + CeB_6 at 2273.0 K and L ↔ δ-Ce + CeB_4 at 1063.0 K), one congruent reaction (L ↔ CeB_6 at 2603.0 K) and one peritectic reaction (L + CeB_6 ↔ CeB_4 at 2438.0 K). Liao et al. [24] re-modified the Ce–B binary system. The temperature of the eutectic reaction (L ↔ β-B + CeB_6) was recommended to be 2263 K. Recently, using the differential thermal analysis (DTA), electron probe microanalysis (EPMA) and X-ray diffraction (XRD), the liquidus in the Ce-rich part and two invariant reactions (L ↔ δ-Ce + CeB_4 and δ-Ce + CeB_4 ↔ γ-Ce) were measured experimentally by Premovic et al. [22]. The solubility of B in δ-Ce and γ-Ce or Ce in β-B was not reported in the literature. Table 1 shows the crystallographic information of the stable phases in the Ce–B binary system.

The enthalpy increment ($H(T) - H(298)$) of the intermetallic compound CeB_6 was measured by Muratov et al. [26] using the vacuum calorimetric equipment with double thin-walled tantalum vessels in the wide temperature range (450–2300 K). The heat capacity of CeB_6 was calculated based on the experimental enthalpy increment [26]. Bolgar et al. [27] determined the enthalpy increment ($H(T) - H(298)$) of the intermetallic compound CeB_4 using the drop calorimetry method in the temperature range of 400–2300 K. An approximate equation to express the temperature dependence of heat capacity and enthalpy increment ($H(T) - H(298)$) of CeB_4 from 298 K to its melting point was obtained. On the other hand, the enthalpy of formation of CeB_6 was measured by Samsonov and Grodstein [28] and Ames and McGrath [29]. Both Samsonov and Grodstein [28] and Ames and McGrath [29] used mass spectrometry method. Moreover, the enthalpy of formation of CeB_4 was measured by Meschel and Kleppa [30] using the direct synthesis calorimetry. Using density functional theory, the enthalpies of formation of CeB_4 and CeB_6 were calculated by the Materials Project [31] and the Open Quantum Database (OQMD) [32]. Table 2 summarizes information on thermodynamic properties of intermetallic compounds, while Table 3 shows phase equilibria mentioned above in the Ce–B binary system.

Table 2
The experimental and calculated enthalpies of formation of intermetallic compounds in the RE-B (RE = Ce, Dy, Lu) binary systems.

Compounds	Enthalpy of formation (kJ/mol-atom)	Method	References
CeB_4	−53.620	Calculation	[21]
	−51.339	Calculation	[22]
	−53.5 ± 1.5	First-principle calculation	[30]
	−60.385	First-principle calculation	[31]
	−49.904	Experiment	[32]
	−56.288	Optimization	This work
CeB_6	−50.541	Calculation	[21]
	−46.457	Calculation	[22]
	−48.4 ± 9.6	Experiment	[28]
	−30.6 ± 27	Experiment	[29]
	−49.712	First-principle calculation	[31]
	−47.019	First-principle calculation	[32]
	−48.531	Optimization	This work
DyB_2	−30.2 ± 1.3	Experiments	[37]
	−54.440	Experiments	[38]
	−54.135	First-principle calculation	[30]
	−54.038	First-principle calculation	[31]
	−58.019	Optimization	This work
DyB_4	−57.692	First-principle calculation	[30]
	−57.596	First-principle calculation	[31]
	−55.007	Optimization	This work
DyB_6	−38.558	First-principle calculation	[31]
	−38.940	Optimization	This work
DyB_{12}	24.038	First-principle calculation	[30]
	−24.212	Optimization	This work
DyB_{66}	−4.686	Optimization	This work
LuB_2	−66.900	First-principle calculation	[38]
	−66.538	First-principle calculation	[30]
	−66.249	First-principle calculation	[31]
	−65.781	Optimization	This work
LuB_4	−53.700	First-principle calculation	[38]
	−53.750	First-principle calculation	[30]
	−53.558	First-principle calculation	[31]
	−56.878	Optimization	This work
LuB_{12}	−23.700	First-principle calculation	[38]
	−23.654	First-principle calculation	[30]
	−23.654	First-principle calculation	[31]
	−24.605	Optimization	This work
LuB_{66}	−6.059	Optimization	This work

Table 3
Information for phase equilibria and thermodynamic properties of in the RE-B (RE = Ce, Dy, Lu) binary systems.

Systems	Information	Method	References
Ce–B	phase equilibria	DTA, XRD	[23]
		DTA, XRD, EPMA	[22]
	enthalpy of formation of compounds	mass spectrometry	[28,29]
		direct synthesis calorimetry	[30]
		first-principle calculation	[31,32]
	heat capacity of compounds	calorimetry	[26]
	enthalpy increment of compounds	calorimetry	[26]
		drop calorimetry	[27]
Dy–B	phase equilibria	optical pyrometer	[33]
		optical pyrometer	[34]
	enthalpy of formation of compounds	direct synthesis calorimetry	[37]
		first-principle calculation	[31,32, 38]
Lu–B	phase equilibria	XRD, electron microscope	[23]
	enthalpy of formation of compounds	first-principle calculation	[31,32, 38]

2.2. The Dy–B binary system

The melting temperature of DyB_6 was determined by Mordovin and Timofeeva [33] to be 2473 K using optical pyrometer, while the melting temperature of DyB_{66} was measured to be 2423 ± 100 K by Spear and Solovyev [34] using optical pyrometer. According to the experimental results [33,34], Spear [23] reviewed the Dy–B binary system. The temperatures of four peritectic reactions (L + DyB_4 ↔ DyB_6, L + DyB_6 ↔ DyB_{12}, L + DyB_4 ↔ DyB_2, L + DyB_{12} ↔ DyB_{66}) were recommended to be 2473.0 K, 2373.0 K, 2373.0 K and 2298 ± 50 K, respectively, while there are two eutectic reactions (L ↔ DyB_2 + α-Dy at 1673.0 K and L ↔ DyB_{66} + β-B) and one congruent reaction (L ↔ DyB_4 at 2773.0 K) in the Dy–B binary system. The crystallographic information of stable phases in the Dy–B binary system was reported by Novikov et al. [35] and Etourneau et al. [36] as given in Table 1, while phase equilibiria information was shown in Table 3.

As for thermodynamic properties of the Dy–B binary system, the enthalpy of formation of DyB_2 was measured by Meschel and Kleppa [37] using direct synthesis calorimetry. Using the first principle calculation, Colinet and Tedenac [38] calculated the enthalpy of formation of DyB_2. The calculated enthalpies of formation of three intermediate compounds, DyB_2, DyB_4, DyB_6 and DyB_{12} by the Materials Project [31] and the Open Quantum Database (OQMD) [32]. Table 2 gives the enthalpies of formation of intermetallic compounds in the Dy–B binary system.

2.3. The Lu–B binary system

Phase equilibria of the Lu–B binary system was studied experimentally by Spear [23] through XRD and electron microscope. Spear [39] compiled the Dy–B binary phase diagram. There are two solid solution phases (α-Lu and β-B) and four intermetallic compounds (LuB_2, LuB_4, LuB_{12} and LuB_{66}) in the Lu–B binary system. Two eutectic reactions (L ↔ LuB_2 + α-Lu at 1773.0 K and L ↔ LuB_{66} + β-B at 2333.0 K) and three peritectic reactions (L + LuB_4 ↔ LuB_2 at 2523.0 K, L + LuB_4 ↔ LuB_{12} at 2449.0 K and L + LuB_{12} ↔ LuB_{66} at 2373.0 K) were identified. LuB_4 melts congruently at 2873.0 K. The solubility of Lu in β-B or B in α-Lu was not reported in the literature. The crystallographic information of the solid solution phases and intermetallic compounds was reviewed by Etourneau et al. [36] as shown in Table 1.

According to the published literature, thermodynamic properties of the Lu–B binary system were not determined experimentally. Using the first principle calculation, the enthalpies of formation of three intermediate compounds, LuB_2, LuB_4 and LuB_{12}, were calculated by Colinet and Tedenac [39]. Moreover, the enthalpies of formation of LuB_2, LuB_4 and LuB_{12} were calculated by the Materials Project [31] and the Open Quantum Database (OQMD) [32]. Table 2 shows the calculated enthalpy of formation of intermetallic compounds in the Lu–B binary system.

3. Thermodynamic modeling

3.1. Pure elements

The Gibbs energy function for the pure element i (i = Ce, Dy, Lu and B) in the φ phase is described by the following equation:

$$^{\circ}G_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \quad (1)$$

where H_i^{SER} is the molar enthalpy of element i at 298.15 K and 1 bar in its standard element reference (SER), while R is the gas constant and T is the absolute temperature in Kelvin. In this work, the Gibbs energies for RE (RE = Ce, Dy, Lu) and B are taken from the Scientific Group Thermodata Europe (SGTE) compiled by Dinsdale [40].

3.2. Solid solution phases

The solid solution phases φ (φ = liquid, Fcc_A1, Bcc_A2, Hcp_A3) is described with the substitutional solution model. The molar Gibbs energy of the phase φ is expressed:

$$G_m^{\phi} = x_B{}^{\circ}G_B^{\phi} + x_{RE}{}^{\circ}G_{RE}^{\phi} + RT(x_B\ln x_B + x_{RE}\ln x_{RE}) + {}^{ex}G_m^{\phi} \quad (2)$$

$$^{ex}G_m^{\phi} = x_B x_{RE}\sum_{j=0}^{n}(A_j + B_jT)(x_B - x_{RE})^j \quad (3)$$

where R is the gas constant, x_B and x_{RE} are molar fractions of the elements B and RE (RE = Ce, Dy, Lu), A_j and B_j are model parameters to be optimized in this work.

3.3. Stoichiometric compounds

In the RE-B (RE = Ce, Dy, Lu) binary systems, the intermetallic compounds, CeB_4, CeB_6, DyB_2, DyB_4, DyB_6, DyB_{12}, DyB_{66}, LuB_2, LuB_4, LuB_{12} and LuB_{66} are treated as stoichiometric compounds. The heat capacities of two intermetallic compounds, CeB_4 and CeB_6, were determined experimentally by Muratov et al. [26] and Bolgar et al. [27]. The molar Gibbs energies ($G_m^{CeB_4}$ and $G_m^{CeB_6}$) of CeB_4 and CeB_6 are expressed as:

$$G_m^{CeB_4} = C_1 + D_1T + E_1T\ln(T) + F_1T^2 + G_1T^3 + H_1T^{-1} + 0.2G_{Ce}^{bcc} + 0.8G_B^{beta} \quad (4)$$

$$G_m^{CeB_6} = C_2 + D_2T + E_2T\ln(T) + F_2T^2 + G_2T^3 + H_2T^{-1} + 0.1429G_{Ce}^{bcc} + 0.8571G_B^{beta} \quad (5)$$

The parameters C_i, D_i, E_i, F_i, G_i, H_i, are to be optimized in this work.

On the other hand, due to the lack of experimental heat capacity of DyB_2, DyB_4, DyB_6, DyB_{12}, DyB_{66}, LuB_2, LuB_4, LuB_{12} and LuB_{66}, their molar Gibbs energies are expressed as:

$$G_m^{RE_pB_q} = \frac{p}{p+q}{}^{\circ}G_{RE}^{\phi} + \frac{q}{p+q}{}^{\circ}G_B^{beta} + J + QT \quad (6)$$

where p and q are the values of stoichiometric number, and the parameters J and Q are to be optimized in this work.

Table 4
Thermodynamic parameters of the Ce–B binary system.

Phase	Thermodynamic Parameters	Reference
Liquid	${}^{0}L^{liq}_{B,Ce} = -63630 - 20T$ ${}^{1}L^{liq}_{B,Ce} = -52581 - 10T$ ${}^{2}L^{liq}_{B,Ce} = -46297 - 5T$ ${}^{3}L^{liq}_{B,Ce} = -25294$	This work
bcc (δ-Ce)	${}^{0}G^{bcc}_{Ce}$ cited from SGTE database	[40]
fcc (γ-Ce)	${}^{0}G^{fcc}_{Ce}$ cited from SGTE database	[40]
rhombohedral (β-B)	${}^{0}G^{rhomb}_{B}$ cited from SGTE database	[40]
CeB_4	$G^{CeB_4}_m = -55000 - 6.92507T + 1.4T\ln T + 0.001T^2 - 3.5\times10^{-7} - 140000T^{-1}$ $+0.8\,{}^{0}G^{rhombo}_{B} + 0.2\,{}^{0}G^{fcc}_{Ce}$	This work
CeB_6	$G^{CeB_6}_m = -47944 + 0.25575T - 0.006T\ln T + 0.002T^2 - 4.1\times10^{-7}T^3 - 80000T^{-1}$ $+0.1429\,{}^{0}G^{fcc}_{Ce} + 0.8571\,{}^{0}G^{rhomb}_{B}$	This work

Table 5
Thermodynamic parameters of the Dy–B binary system.

Phase	Thermodynamic Parameters	Reference
Liquid	${}^{0}L^{liq}_{B,Dy} = -81097 + 2.927T$ ${}^{1}L^{liq}_{B,Dy} = -39004 - 3.93T$ ${}^{2}L^{liq}_{B,Dy} = -10000$	This work
hcp (α-Dy)	${}^{0}G^{hcp}_{Dy}$ cited from SGTE database	[40]
bcc (β-Dy)	${}^{0}G^{bcc}_{Dy}$ cited from SGTE database	[40]
rhombohedral (β-B)	${}^{0}G^{rhomb}_{B}$ cited from SGTE database	[40]
DyB_2	$G^{DyB_2}_m = -58011.6 + 8.521793T + 0.3333\,{}^{0}G^{hcp}_{Dy} + 0.6667\,{}^{0}G^{rhomb}_{B}$	This work
DyB_4	$G^{DyB_4}_m = -55011 + 6.3634T + 0.2\,{}^{0}G^{hcp}_{Dy} + 0.8\,{}^{0}G^{rhomb}_{B}$	This work
DyB_6	$G^{DyB_6}_m = -40861 + 4.709T + 0.1429\,{}^{0}G^{hcp}_{Dy} + 0.8571\,{}^{0}G^{rhomb}_{B}$	This work
DyB_{12}	$G^{DyB_{12}}_m = -24104.769 + 3.21794T + 0.0769\,{}^{0}G^{hcp}_{Dy} + 0.9231\,{}^{0}G^{rhomb}_{B}$	This work
DyB_{66}	$G^{DyB_{66}}_m = -5000.5 + 0.462333T + 0.0149\,{}^{0}G^{hcp}_{Dy} + 0.9851\,{}^{0}G^{rhomb}_{B}$	This work

Table 6
Thermodynamic parameters of the Lu–B binary system.

Phase	Thermodynamic Parameters	Reference
Liquid	${}^{0}L^{liq}_{B,Lu} = -100001 + 9.6885T\,{}^{1}L^{liq}_{B,Lu} = -38485 + 2.2186T$	This work
hcp (α-Lu)	${}^{0}G^{hcp}_{Lu}$ cited from SGTE database	[40]
rhombohedral (β-B)	${}^{0}G^{rhomb}_{B}$ cited from SGTE database	[40]
LuB_2	$G^{LuB_2}_m = -66005 + 10.7237T + 0.3333\,{}^{0}G^{hcp}_{Lu} + 0.6667\,{}^{0}G^{rhomb}_{B}$	This work
LuB_4	$G^{LuB_4}_m = -55000 + 6.17T + 0.2\,{}^{0}G^{hcp}_{Lu} + 0.8\,{}^{0}G^{rhomb}_{B}$	This work
LuB_{12}	$G^{LuB_{12}}_m = -25005 + 3.2908T + 0.0769\,{}^{0}G^{hcp}_{Lu} + 0.9231\,{}^{0}G^{rhomb}_{B}$	This work
LuB_{66}	$G^{LuB_{66}}_m = -5822 + 0.76952T + 0.0149\,{}^{0}G^{hcp}_{Lu} + 0.9581\,{}^{0}G^{rhomb}_{B}$	This work

4. Results and discussion

Based on the experimental information mentioned above, thermodynamic parameters of all phases in the RE-B (RE = Ce, Dy, Lu) binary systems were optimized using the module PARROT of the Thermo-Calc® software. The PARROT module works through minimizing the square sum of the differences between the experimental data and the calculated values. During the optimization, each set of experimental data is given a certain weight according to the reliability and compatibility of experimental data. In this work, thermodynamic parameters of the RE-B (RE = Ce, Dy, Lu) binary systems were obtained finally as given in Table 4, Table 5 and Table 6, respectively. The comparison of the calculated results with the experimental data was discussed in the following sections.

4.1. The Ce–B binary system

Fig. 1 compares the calculated Ce–B binary phase diagram in this work with the calculated results [21,22]. Fig. 2 is the calculated phase diagram with the experimental data measured by Spear [23] and Premovic et al. [22]. The calculated phase boundary of liquid phase in the Ce-rich part agrees well with the experimental results [22,23]. Table 7 gives the calculated temperatures and compositions of invariant reactions in the Ce–B binary system with the experimental data [22,23] and the calculated results [21,22]. In this work, the temperatures of two eutectic reactions, L ↔ δ-Ce + CeB_4 and L ↔ β-B + CeB_6, were calculated to be 1060.3 K and 2278.4 K, respectively, while the calculated temperatures of the peritectic reaction, L + CeB_6 ↔ CeB_4, and the congruent reaction, L ↔ CeB_6, is 2446.1 K and 2603.1 K. The temperature of the peritectoid reaction, δ-Ce + CeB_4 ↔ γ-Ce, was calculated to be 999.8 K. As can be seen in Table 6, the calculated temperatures and compositions of invariant reactions in the Ce–B binary system in this work are well consistent with the experimental data measured by Premovic et al. [22] and Spear [23].

Fig. 3 shows the calculated enthalpies of formation of intermetallic compounds in the Ce–B binary system at 298 K with the experimental data [28–30] and the calculated results [21,22,31,32]. In this work, the calculated enthalpy of formation of CeB_4 is in good agreement with the experimental data [30] and the calculated results [31,32], while the calculated enthalpy of formation of CeB_6 is obvious difference with the

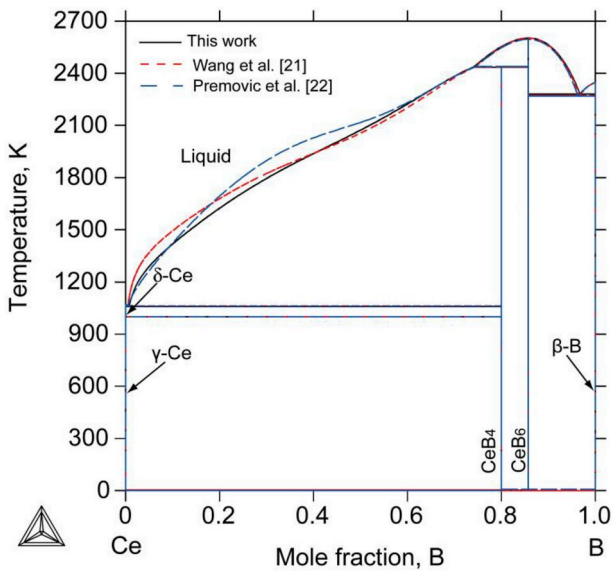

Fig. 1. The calculated Ce–B phase diagram in this work with the calculated results [21,22].

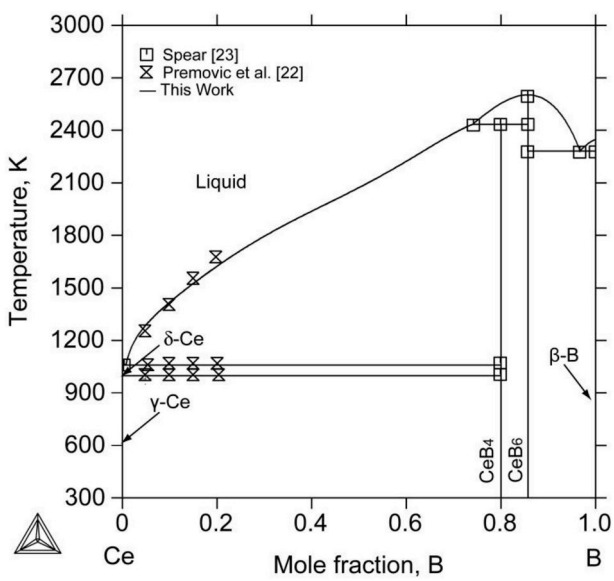

Fig. 2. The calculated Ce–B phase diagram in this work with the experimental data [22,23].

experimental data [29]. It should be pointed out that the experimental data [29] show large experimental error and are less reliable. The enthalpy of formation of the CeB_6 compounds should be still measured in the further experimental investigations.

Fig. 4 is the calculated heat capacities of CeB_4 and CeB_6 in this work with the experimental data measured by Muratov et al. [26] and Bolgar et al. [27] as well as the calculated results by Wang et al. [21] and Premovic et al. [22]. The calculated heat capacities of these two compounds in this work are well consistent with the experimental data [26,27]. Fig. 5 shows the calculated enthalpy increments of CeB_4 and CeB_6 in this work. The calculated results in this work are well consistent with the experimental data [26,27] and the calculated results [22]. Fig. 6 is the calculated enthalpy of mixing of liquid Ce–B alloy at 3000 K in this work and the calculated results in Refs. [21,22] and by the Miedema's theory [43,44]. As can be seen, the calculated results by Premovic et al. [22] show an abnormally positive deviation in the Ce-rich corner, which is

Table 7
Invariant reactions in the Ce–B binary system.

Reactions	Reaction Type	Composition (x_B^L)	T(K)	Reference
L ↔ CeB_6	Congruent	0.857	2603.0	[23]
		0.857	2603.0	[24]
		0.857	2603.0	[21]
		0.857	2604.0	[22]
		0.857	2603.1	This work
L + CeB_6 ↔ CeB_4	Peritectic	0.740	2438.0	[23]
		0.740	2438.0	[24]
		0.744	2237.8	[21]
		0.741	2433.8	[22]
		0.740	2435.0	This work
L ↔ CeB_6 + β-B	Eutectic	0.990	2273.0	[23]
		0.990	2263.0	[24]
		0.967	2273.0	[21]
		0.964	2270.7	[22]
		0.967	2278.4	This work
L ↔ δ-Ce + CeB_4	Eutectic	0.010	1063.0	[23]
		0.010	1063.0	[24]
		0.010	1061.8	[21]
		0.006	1063.3	[22]
		0.006	1058.3	This work
δ-Ce + CeB_4 ↔ γ-Ce	Eutectoid	–	999.0	[23]
		–	999.0	[24]
		–	999.8	[21]
		–	999.9	[22]
		–	999.8	This work

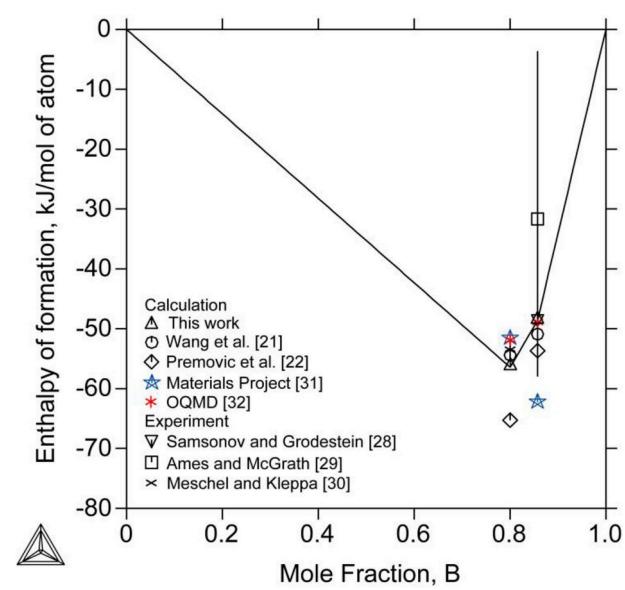

Fig. 3. The calculated enthalpies of formation of Ce–B intermetallic compounds at 298 K with the experimental data [28–30] and the calculated results [21,22,31,32]. (Reference states: γ-Ce and β-B).

obvious deviation with the calculated results by the Miedema model [43,44]. The tendency of the calculated enthalpy of mixing of liquid phase in this work is consistent with the calculated results by the Miedema's theory [43,44], although the calculated enthalpy of mixing of liquid phase in this work is much more positive than the calculated results by the Miedema's theory [43,44], the tendency of mixing enthalpy is similar.

4.2. The Dy–B binary system

Fig. 7 is the calculated Dy–B binary phase diagram with the

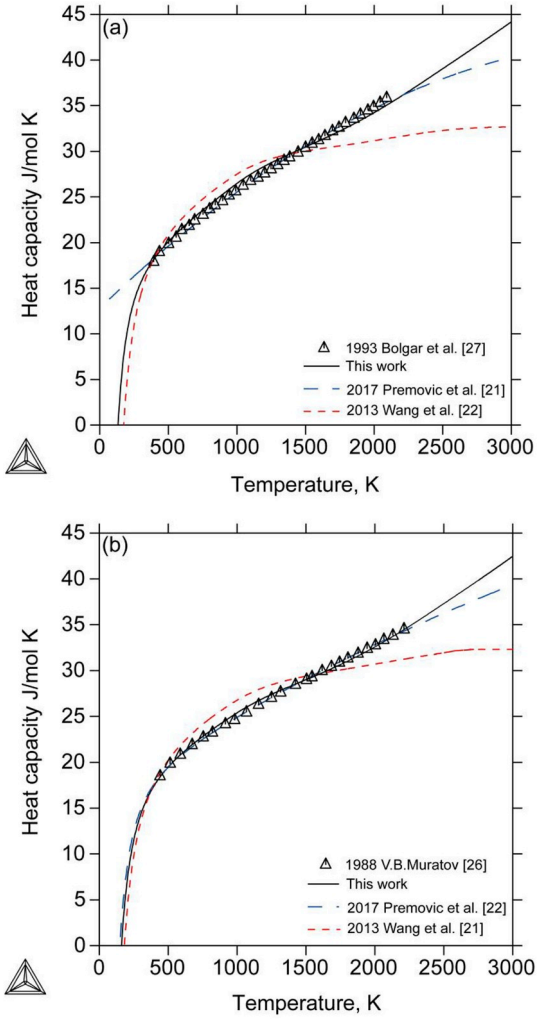

Fig. 4. Heat capacity of the intermetallic compounds CeB_4 (a) and CeB_6 (b) with the experimental data [26,27] and the calculated results [21,22].

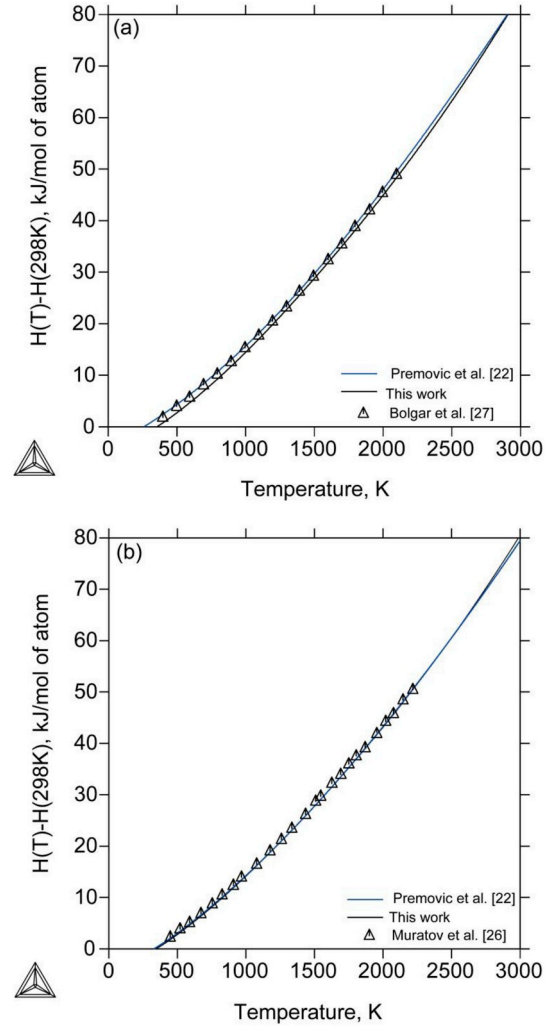

Fig. 5. Enthalpy increment of the intermetallic compounds CeB_4 (a) and CeB_6 (b) with the experimental data and [26,27] the calculated results [22].

experimental data [33,34] and the assessed results [23]. As can be seen, five intermetallic compounds, DyB_2, DyB_4, DyB_6, DyB_{12} and DyB_{66}, are stable at room temperature in this work. Table 8 compares the calculated temperatures and compositions of invariant reactions in the Dy–B binary system in this work with the experimental data [33,34] and the calculated results [25] and the assessed results [23]. The calculated temperatures of two eutectic reactions, L ↔ DyB_{66} + β-B and L ↔ DyB_2 + α-Dy, are 2346.1 K and 1614.0 K, respectively, while the calculated temperatures of five peritectic reactions, L + DyB_4 ↔ DyB_6, L + DyB_6 ↔ DyB_{12}, L + DyB_{12} ↔ DyB_{66}, L + DyB_4 ↔ DyB_2 and L + β-Dy ↔ α-Dy, are 2472.0 K, 2370.3 K, 2349.9 K, 2375.0 K and 1639.7 K. The calculated temperatures and compositions of invariant reactions in the Dy–B binary diagram are in good agreement with the literature data [23,33,34].

Fig. 8 shows the calculated enthalpies of formation of intermetallic compounds in the Dy–B binary system at 298 K. As can be seen, the calculated enthalpy of formation of DyB_2 in this work is in good agreement with the calculated results [31,32,38], and is much more negative than the experimental data determined by Meschel and Kleppa [37] and the assessed results [25]. According to the reported experimental information, the enthalpies of formation of REB_2 intermetallic compounds determined by Meschel and Kleppa [37] in the RE-B binary systems were found to be systematically more positive than the calculated results [31,32]. It indicates that the enthalpies of formation of REB_2 compounds should be the most negative in the RE-B binary systems. Therefore, the calculated enthalpies of formation of intermetallic

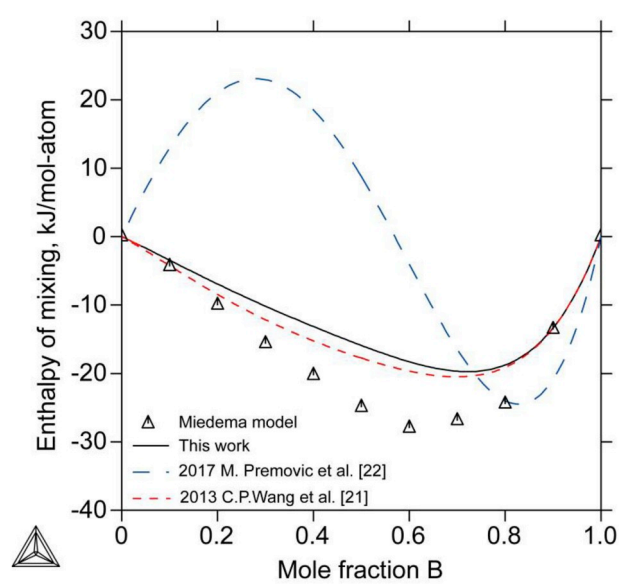

Fig. 6. The calculated enthalpy of mixing of liquid Ce–B alloy at 3000 K in this work with calculated results [21-22] and calculated by Miedema model. (Reference states: liquid Ce, liquid B).

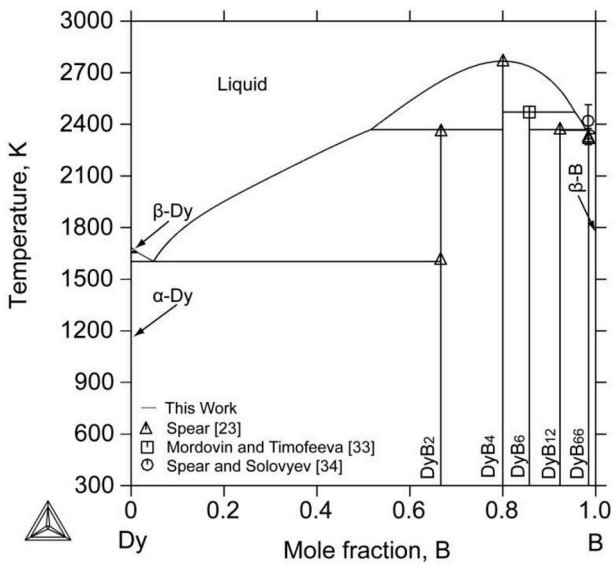

Fig. 7. Calculated Dy–B phase diagram with the experimental data [33,34] and the assessed results [23].

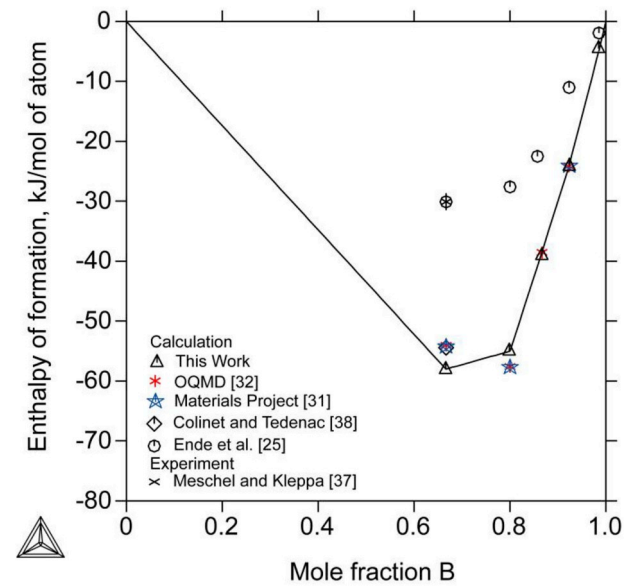

Fig. 8. Calculated enthalpy of formation of Dy–B intermetallic compounds at 298 K with the experimental data [37] and the calculated results [25,31,32,38]. (Reference states: α-Dy and β-B).

Table 8
Invariant reactions in the Dy–B binary system.

Reactions	Reaction Type	Composition (x_B^L)	T(K)	Reference
L ↔ DyB_4	Congruent	0.800	2773.0	[23]
		0.800	2749.3	[25]
		0.800	2767.1	This work
L + DyB_4 ↔ DyB_6	Peritectic	0.880	2473.0	[23]
		–	2475.9	[25]
		–	2472.0	[33]
		0.955	2472.0	This work
L + DyB_6 ↔ DyB_{12}	Peritectic	0.930	2373.0	[23]
		–	2378.8	[25]
		0.978	2371.3	This work
L + DyB_{12} ↔ DyB_{66}	Peritectic	–	2298.0 ± 50	[23]
		–	2423.0 ± 100	[34]
		–	2347.0	[25]
		0.979	2365.0	This work
L + DyB_4 ↔ DyB_2	Peritectic	0.620	2373.0	[23]
		–	2364.1	[25]
		0.515	2369.8	This work
L ↔ DyB_{66} + β-B	Eutectic	0.990	–	[23]
		0.995	2343.0	[25]
		0.998	2346.1	This work
L+β-Dy ↔α-Dy	Peritectic	–	1654.0	[25]
		0.028	1639.7	This work
L ↔ DyB_2 + α-Dy	Eutectic	–	1673.0	[23]
		0.039	1619.1	[25]
		0.048	1608.7	This work

compounds in the Dy–B binary system in this work are acceptable and still need to be measured in the further experimental investigations.

4.3. The Lu–B binary system

Fig. 9 is the calculated Lu–B binary phase diagram in this work with the experimental data [23]. Four intermetallic compounds, LuB_2, LuB_4, LuB_{12} and LuB_{66}, are stable at room temperature in this work. The calculated temperatures of two eutectic reactions (L ↔ LuB_{66} + β-B and L ↔ LuB_2 + α-Lu) are 2348.7 K and 1768.5 K, while the calculated

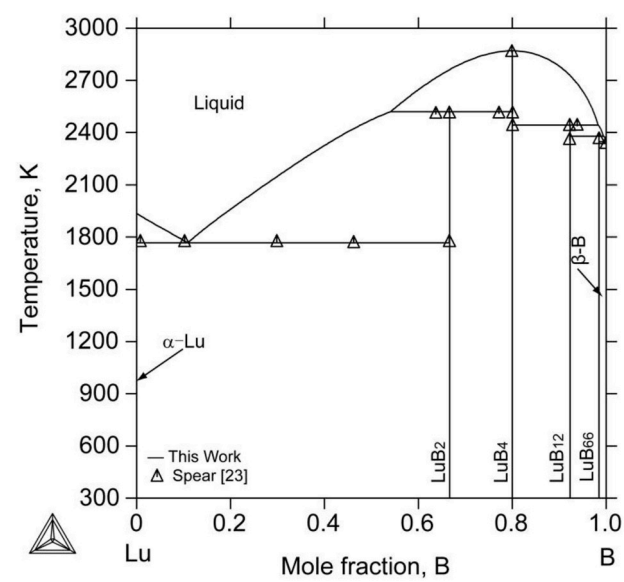

Fig. 9. Calculated Lu–B binary phase diagram with the experimental data [23].

Table 9
Invariant reactions in the Lu–B binary system.

Reactions	Reaction Type	Composition (x_B^L)	T(K)	Reference
L ↔ LuB_4	Congruent	0.800	2873.0	[23]
		0.800	2872.9	This work
L + LuB_4 ↔ LuB_{12}	Peritectic	0.930	2449.0	[23]
		0.983	2445.8	This work
L + LuB_{12} ↔ LuB_{66}	Peritectic	0.980	2373.0	[23]
		0.994	2377.3	This work
L + LuB_4 ↔ LuB_2	Peritectic	0.620	2523.0	[23]
		0.541	2525.1	This work
L ↔ LuB_{66} + β-B	Eutectic	0.990	2333.0	[23]
		0.999	2348.7	This work
L ↔ LuB_2 + α-Lu	Eutectic	0.100	1773.0	[23]
		0.109	1768.5	This work

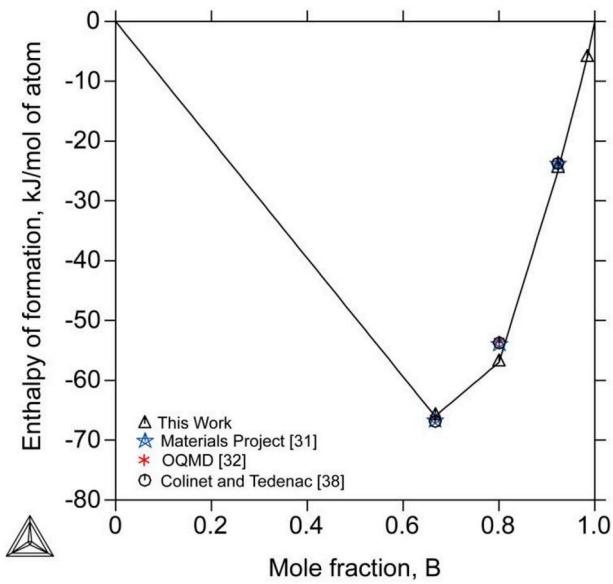

Fig. 10. Calculated enthalpy of formation of Lu–B intermetallic compounds at 298 K with the calculated results [31,32,38]. (Reference states: α-Lu and β-B).

temperatures of three peritectic reactions ($L + LuB_4 \leftrightarrow LuB_{12}$, $L + LuB_{12} \leftrightarrow LuB_{66}$ and $L + LuB_4 \leftrightarrow LuB_2$) are 2449.0 K, 2377.3 K and 2525.1 K, respectively. The congruent reaction, $L \leftrightarrow LuB_4$, is calculated to be 2873.3 K. Table 9 shows the calculated temperatures and compositions of invariant reactions in the Lu–B binary system with the experimental data [23].

Fig. 10 is calculated the enthalpy of formation of intermetallic compounds in the Lu–B binary system at 298 K with the calculated results [31,32,38]. In this work, the calculated enthalpies of formation of LuB_2, LuB_4 and LuB_{12} are in good agreement with the calculated results [31,32,38].

Considering the similarity in the RE-B binary systems, the experimental and calculated enthalpy of formation of intermetallic compounds in RE-B (RE = Ce, Dy, Lu) systems were summarized in Table 2. As can be seen, the enthalpies of formation of RE-B compounds become more and more negative with the increase of RE atomic number, and the enthalpy of formation of REB_2 intermetallic compound is most negative. Moreover, the calculated results in this work are in good agreement with the experiment data [28,30] and the calculated results [31,32,38], and thus are acceptable and reasonable. It should be pointed out that the further experiments should be still necessary to measure accurately thermodynamic properties of the RE-B binary systems.

5. Conclusions

In this work, based on the experimental data including phase equilibria and thermodynamic data reported in the literature, the RE-B (RE = Ce, Dy, Lu) binary systems were assessed thermodynamically using the CALPHAD method. Self-consistent thermodynamic parameters of the RE-B (RE = Ce, Dy, Lu) binary systems were obtained finally, which can reproduce well most of the experimental data including phase equilibria and thermodynamic properties. It is indispensable to develop the compatible and available thermodynamic database of the multi-component RE-Fe-B magnetic alloys.

Acknowledgements

This work was supported financially by the National Key Research and Development Program of China (2016YFB0700901), the National Basic Foundation of China (2014CB643703), the National Natural Science Foundation of China (51761008, 51971069, 51461013), the Guangxi Natural Science Foundation (2016GXNSFDA380015, 2016GXNSFGA380001), the Guangxi Project of Science and Technology (2017AD23031) and Guangxi Key Laboratory of Information Materials, Guilin University of Electronic Technology, China (171005-Z).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2020.101740.

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